

D854

Environmental Science

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

A town of 18,000 people draws its water from an aquifer recharged by about 4 million cubic meters of rain each year. The town pumps 6 million cubic meters a year. Its water utility recently won a regional customer-service award. Which statement is correct?

- A. Groundwater is renewable, so any pumping rate is sustainable.
- B. The aquifer is being depleted by about 2 million m³ a year, so current use is unsustainable even though groundwater is renewable.
- C. Use is sustainable because the utility is well run.
- D. The aquifer gains about 2 million m³ a year.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2.

Twelve independent fishing companies share an open-access bay with no catch limits. Each company knows the cod stock is shrinking but reasons that any fish it leaves will just be caught by others. The bay also hosts a popular summer regatta. Which policy most directly addresses the underlying problem?

- A. Set an enforced total catch limit and divide it into individual quotas for each company.
- B. Ask each company to fish less voluntarily.
- C. Move the regatta to reduce stress on the cod.
- D. Subsidize larger boats so companies can fish farther out.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

A lake beside a golf course with 18 irrigated fairways receives fertilizer runoff each spring. By late summer a thick algal bloom forms and then dies back, and dead fish are found near the bottom. The course recently switched to electric carts. What most directly killed the fish?

- A. The bloom raised the water's pH until fish could not survive.
- B. The algae produced too much oxygen for the fish.
- C. The fertilizer is directly toxic to the fish.
- D. Bacteria decomposing the dead algae used up the dissolved oxygen.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4. [TRANSFER — reversed stem / different variable]

In a lake with a heavy algal bloom, when is dissolved oxygen in the surface water usually lowest?

- A.** At noon, when algae are most active.
- B.** Mid-afternoon.
- C.** Just before dawn.
- D.** Oxygen stays constant throughout the day.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5.

A river running through a county with three state parks receives pollution from a paper mill's discharge pipe and from runoff across thousands of acres of farmland. The mill employs 400 people. Under a typical permit system, which source is easier to regulate directly?

- A.** Neither, because the mill employs too many people to regulate.
- B.** The paper mill, because its pipe is a point source that can be permitted and monitored.
- C.** Neither — both a pipe and field runoff count as nonpoint sources.
- D.** The farmland, because it covers a larger area.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6. [TRANSFER — reversed stem / different variable]

Which is an example of a nonpoint source of water pollution?

- A.** A leaking underground tank at one gas station.
- B.** Oil and road salt washing off city streets during rain.
- C.** A factory pipe that discharges only at night.
- D.** A sewage treatment plant's outfall pipe.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

On hot, sunny afternoons in a valley city with a growing tech sector, ground-level ozone peaks, though no factory emits ozone directly. Morning commuter traffic is heavy. The city recently hosted a marathon. What best explains the afternoon ozone?

- A. Ozone forms mainly at night when the air is still.
- B. Ozone is released directly from car tailpipes.
- C. Ozone leaks down from the stratospheric ozone layer.
- D. Ozone is a secondary pollutant formed when nitrogen oxides and VOCs from sources such as vehicles react in sunlight.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8. [TRANSFER — reversed stem / different variable]

Which of these is a primary air pollutant?

- A. Ground-level ozone.
- B. Photochemical smog as a whole.
- C. Carbon monoxide emitted from vehicle exhaust.
- D. Sulfuric acid in acid rain.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9.

A farm on rolling hills, owned by the same family for four generations, loses topsoil every spring when heavy rain runs straight down its plowed rows. The farm sells corn to a regional ethanol plant. Which practice most directly reduces this erosion?

- A. Stop selling corn to the ethanol plant.
- B. Add more fertilizer to replace the lost nutrients.
- C. Plow rows straight downhill so water drains faster.
- D. Plow and plant along the contours of the slope, and keep a cover crop on the soil between seasons.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10.

An old fuel depot covering 5 hectares has soil with moderate petroleum hydrocarbon contamination in the top meter. The city's budget is tight, and the depot's historic brick office is listed on a local register. Which approach can break down the contamination on site at relatively low cost?

- A.** In-situ bioremediation: add nutrients and oxygen so soil microbes break down the hydrocarbons.
- B.** Excavate and landfill all soil, because removal is always the best option.
- C.** Cap the site with asphalt, which removes the hydrocarbons.
- D.** Leave the site untouched to protect the historic office.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

A coastal region of 2 million people, famous for its seafood festivals, is growing at 3.5% per year. A regional water board wants to know when water demand will double if use per person stays the same. About how many years will that take?

- A.** 20 years
- B.** 35 years
- C.** 0.05 years
- D.** 28.6 years

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12. [TRANSFER — reversed stem / different variable]

A city's population doubles in 14 years. What is its approximate annual growth rate?

- A.** 14%
- B.** 7.1%
- C.** 5%
- D.** 0.2%

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer B Concept 1

A town of 18,000 people draws its water from an aquifer recharged by about 4 million cubic meters of rain each year. The town pumps 6 million cubic meters a year. ~~Its water utility recently won a regional customer service award.~~ Which statement is correct?

Target word: sustainability of the pumping **Decides:** Withdrawal exceeds recharge by 2 million m³ per year.

A	Absolute language	Renewable resources are sustainable only at or below their renewal rate.
B	Correct	Use above the renewal rate draws down storage and lowers the water table.
C	Camouflage	Customer service does not change the water balance.
D	Inverted operation	Subtracts in the wrong direction; pumping exceeds recharge.

2. Answer A Concept 2

Twelve independent fishing companies share an open-access bay with no catch limits. Each company knows the cod stock is shrinking but reasons that any fish it leaves will just be caught by others. ~~The bay also hosts a popular summer regatta.~~ Which policy most directly addresses the underlying problem?

Target word: fix for a commons problem **Decides:** Open access with no enforceable limits.

A	Correct	Enforced shares remove the incentive to race for the fish.
B	Adjacent category	Voluntary restraint leaves the same incentive in place; others can still take what one leaves.
C	Camouflage	The regatta is not the cause of overfishing.
D	Inverted operation	Increases fishing pressure instead of limiting it.

3. Answer D Concept 3

A lake beside a golf course ~~with 18 irrigated fairways~~ receives fertilizer runoff each spring. By late summer a thick algal bloom forms and then dies back, and dead fish are found near the bottom. ~~The course recently switched to electric carts.~~ What most directly killed the fish?

Target word: direct cause of the fish kill **Decides:** Bloom dies back; fish die near the bottom.

A	Adjacent category	pH can shift during blooms, but the fish kill after die-off points to low oxygen.
B	Inverted operation	Living algae add oxygen by day; the die-off removes it.
C	Dropped element	Skips the bloom-decay-oxygen chain; the nutrients act indirectly.
D	Correct	Decomposition consumes oxygen, creating hypoxic bottom water.

4. Answer C Concept 3 • transfer

In a lake with a heavy algal bloom, when is dissolved oxygen in the surface water usually lowest?

Target word: timing of lowest oxygen **Decides:** Photosynthesis stops at night while respiration continues.

A	Confusion set	Algal activity at noon is photosynthesis, which adds oxygen.
B	Inverted operation	Afternoon is when photosynthesis has pushed oxygen highest.
C	Correct	All night, algae and bacteria respire without photosynthesis, so oxygen bottoms out at dawn.
D	Absolute language	Blooms cause large day-night swings.

5. Answer B Concept 4

A river running through a county ~~with three state parks~~ receives pollution from a paper mill's discharge pipe and from runoff across thousands of acres of farmland. ~~The mill employs 400 people.~~ Under a typical permit system, which source is easier to regulate directly?

Target word: which source is easier to regulate **Decides:** A single identifiable discharge pipe.

A	Camouflage	Employment does not change how pollution is classified or permitted.
B	Correct	A single outlet lets regulators set limits and sample directly.
C	Confusion set	A discharge pipe is the textbook point source.
D	Inverted operation	Diffuse runoff over a large area is harder, not easier, to regulate.

6. Answer B Concept 4 • transfer

Which is an example of a nonpoint source of water pollution?

Target word: identify nonpoint **Decides:** Diffuse runoff, no single outlet.

A	Adjacent category	A single identifiable source behaves like a point source.
B	Correct	The pollution enters from countless surfaces, not one outlet.
C	Confusion set	Timing does not change the classification; it is still one pipe.
D	Inverted operation	An outfall pipe is a point source.

7. Answer D Concept 5

On hot, sunny afternoons in a valley city ~~with a growing tech sector~~, ground-level ozone peaks, though no factory emits ozone directly. Morning commuter traffic is heavy. ~~The city recently hosted a marathon.~~ What best explains the afternoon ozone?

Target word: source of ground-level ozone **Decides:** Ozone peaks with sunlight after morning emissions.

A	Inverted operation	Formation needs sunlight; levels peak in the afternoon.
B	Confusion set	Tailpipes emit NO _x and VOCs, not ozone itself.
C	Adjacent category	Stratospheric ozone is a different layer; smog ozone forms near the ground.
D	Correct	Morning emissions plus afternoon sunlight produce the peak.

8. Answer C Concept 5 • transfer

Which of these is a primary air pollutant?

Target word: identify a primary pollutant **Decides:** Emitted directly from the source.

A	Inverted operation	Ozone forms in the air — secondary.
B	Adjacent category	Smog is a mixture dominated by secondary products.
C	Correct	CO comes straight out of the tailpipe.
D	Confusion set	It forms from emitted SO ₂ reacting in the air — secondary.

9. Answer D Concept 6

A farm on rolling hills, ~~owned by the same family for four generations~~, loses topsoil every spring when heavy rain runs straight down its plowed rows. ~~The farm sells corn to a regional ethanol plant.~~ Which practice most directly reduces this erosion?

Target word: erosion control **Decides:** Rows run downhill, channeling runoff.

A	Camouflage	The buyer does not affect how water moves across the field.
B	Adjacent category	Treats a symptom without stopping soil loss.
C	Inverted operation	Faster downhill flow carries more soil away.
D	Correct	Contour rows slow runoff; cover protects the soil surface.

10. Answer A Concept 7

An old fuel depot covering 5 hectares has soil with moderate petroleum hydrocarbon contamination in the top meter. The city's budget is tight, and ~~the depot's historic brick office is listed on a local register.~~ Which approach can break down the contamination on site at relatively low cost?

Target word: remediation method **Decides:** Biodegradable hydrocarbons spread over a large, shallow area.

A	Correct	Petroleum is readily biodegraded, and treating in place avoids hauling costs.
B	Absolute language	Excavation works but is costly and simply moves the waste; "always" is false.
C	Confusion set	Capping contains contamination; it does not remove or destroy it.
D	Camouflage	The listing affects the building, not the need to treat the soil.

11. Answer A Concept 8

A coastal region ~~of 2 million people, famous for its seafood festivals~~, is growing at 3.5% per year. A regional water board wants to know when water demand will double if use per person stays the same. About how many years will that take?

Target word: doubling time **Decides:** $70 \div 3.5$.

A	Correct	$70 \div 3.5 = 20$ years.
B	Wrong stem number	Divides 70 by the 2 (million) population figure.
C	Inverted operation	Divides 3.5 by 70 instead of 70 by 3.5.
D	Confusion set	Uses 100 instead of 70 in the rule.

12. Answer C Concept 8 • transfer

A city's population doubles in 14 years. What is its approximate annual growth rate?

Target word: growth rate from doubling time **Decides:** Rate $\approx 70 \div 14$.

A	Wrong stem number	Reads the doubling time as the rate.
B	Confusion set	Uses 100 instead of 70.
C	Correct	$70 \div 14 = 5\%$ per year.
D	Inverted operation	Divides 14 by 70.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owled.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	1	
Wrong stem number	A figure copied from the prompt that is not the answer.	2	
Adjacent category	The right idea in the wrong bucket.	6	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	11	
Absolute language	Always, never, guarantee — where judgment applies.	3	
Confusion set	Every option is real; the test is exact role-matching.	8	
Camouflage	Answers a stem fact that was never the question.	5	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Sustainable yield	A renewable resource stays sustainable only if use does not exceed its rate of renewal; faster use mines it like a nonrenewable resource.	“Renewable” means it can be used at any rate.	A use rate and a renewal (recharge, growth) rate in the same stem.
2	Tragedy of the commons	Open-access shared resources are overused because each user gains the full benefit of taking more but bears only part of the cost; fixes assign limits or rights that are enforced.	Relying on voluntary restraint alone.	Many users, one shared resource, no enforceable limits.
3	Eutrophication	Excess nutrients cause algal blooms; when algae die, bacterial decomposition uses up dissolved oxygen, causing hypoxia and fish kills.	Thinking fertilizer poisons fish directly or that algae lower oxygen by photosynthesis.	Nutrient runoff followed by a bloom and dead fish.
4	Point vs. nonpoint pollution	A point source discharges from one identifiable spot (a pipe) and is easier to permit and monitor; nonpoint pollution comes from diffuse runoff over wide areas.	Classifying by the size of the polluter instead of how the pollution enters.	A pipe vs. runoff across land.
5	Primary vs. secondary air pollutants	Primary pollutants are emitted directly (CO, SO ₂ , NO _x); secondary pollutants such as ground-level ozone form in the air when primary pollutants react, often in sunlight.	Thinking ozone comes out of tailpipes.	A pollutant that peaks on sunny afternoons with no direct source.
6	Soil conservation	Erosion is reduced by slowing water and keeping soil covered: contour plowing, terracing, cover crops, and no-till.	Treating nutrient loss with fertilizer instead of stopping the erosion.	Rain running straight down plowed slopes.
7	Remediation choices	Treatment (e.g., in-situ bioremediation) destroys or removes a contaminant; containment (capping) only isolates it. Match the method to the contaminant and site.	Believing capping removes the contaminant, or that excavation is always best.	A description of contamination type and extent with a cost constraint.
8	Doubling time (rule of 70)	Doubling time in years $\approx 70 \div$ annual growth rate in percent.	Dividing by the population size or using 100 instead of 70.	A steady percent growth rate and “how long to double.”

Course scope (public description): This course explores the essential concepts and processes related to Earth's natural resources, focusing on the interactions between land, water, and air. Students will analyze how human activities impact these resources and examine the scientific, technological, and policy-driven efforts to manage and remediate environmental issues. Through case studies and real-world examples, the course emphasizes sustainable practices and the role of innovative solutions in addressing global challenges. By the end of the course, students will be equipped to critically evaluate resource management strategies and contribute to environmental

sustainability.